

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ILLUMINA, INC.,
Petitioner,

v.

RAVGEN, INC.,
Patent Owner.

IPR2021-01271
Patent 7,727,720 B2

Before ZHENYU YANG, TIMOTHY G. MAJORS, and DAVID COTTA,
Administrative Patent Judges.

COTTA, *Administrative Patent Judge.*

JUDGMENT

Final Written Decision

Determining No Challenged Claims Unpatentable

35 U.S.C. § 318(a)

Granting Patent Owner's Motion for Entry of Protective Order and to Seal;

Granting Petitioner's Motion to Seal

37 C.F.R. §§ 42.14, 42.54

I. INTRODUCTION

Illumina, Inc. (“Petitioner” or “Illumina”),¹ on July 20, 2021, filed a Petition to institute *inter partes* review of claims 1, 3–9, 21, 22, and 24–30 of U.S. Patent No. 7,727,720 B2 (Ex. 1001, “the ’720 patent”). Paper 1 (“Pet.” or “Petition”). We instituted trial on January 26, 2022. Paper 12 (“Inst. Dec.”). During trial, Ravgen, Inc. (“Patent Owner”)² filed a Patent Owner Response. Paper 19 (“PO Resp.”). Later filings include Petitioner’s Reply (Paper 30 (“Pet. Reply”)) and Patent Owner’s Sur-Reply (Paper 39 (“PO Sur-Reply”)). We also allowed Petitioner a limited Sur-Sur-Reply. Paper 41.³ An oral hearing was held on October 26, 2022, and a transcript is entered in the record. Paper 44 (“Tr.”).

Patent Owner also filed a motion for entry of a protective order and to seal (Paper 20), and Petitioner filed a motion to seal (Paper 28). Both motions were unopposed.

We have jurisdiction under 35 U.S.C. § 6(b). After considering the parties’ arguments and evidence, we determine that Petitioner has not proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

¹ Petitioner identifies itself, Illumina Cambridge, Ltd., and Verinata Health, Inc., as the real parties-in-interest. Pet. 2.

² Patent Owner identifies itself as the real party-in-interest. Paper 4, 1.

³ The permitted scope of this filing is detailed in Exhibit 3007 (*see* email to parties dated October 4, 2022, responding to Petitioner’s request).

A. Related Patents & Proceedings

The '720 patent issued on June 1, 2010, from U.S. Patent Application No. 11/212,812 (“the '812 Application”), filed August 26, 2005. Ex. 1001, codes (21), (22), (45). The '812 Application is a continuation of Application No. PCT/US2004/006337, filed March 1, 2004, which itself claims priority through a chain of several continuation and continuation-in-part applications with filing dates in 2003, and a provisional application filed May 8, 2002. *Id.* at codes (60), (63). Related U.S. Patent No. 7,332,277 (“the '277 patent”) issued on February 19, 2008, and claims priority to common applications. *Id.*; IPR2021-01272, Exhibit 1001 at codes (60), (63).

Petitioner identifies multiple lawsuits involving the '720 patent. Those lawsuits include: *Ravgen, Inc. v. Illumina, Inc.*, No. 20-1644 (D. Del.); *Ravgen, Inc. v. Natera, Inc. and NSTX, Inc.*, No. 1:20-cv-00692-ADA (W.D. Tex.); *Ravgen, Inc. v. Ariosa Diagnostics, Inc. et al.*, C.A. No. 20-1646 (D. Del.); *Ravgen, Inc. v. Myriad Genetics, Inc. et al.*, C.A. No. 20-1730 (D. Del.); *Ravgen, Inc. v. Progenity, Inc.*, C.A. No. 20-1734 (D. Del.); *Ravgen, Inc. v. Natera, Inc. et al.*, C.A. No. 20-451 (W.D. Tex.); *Ravgen, Inc. v. PerkinElmer, Inc. et al.*, C.A. No. 20-452 (W.D. Tex.); *Ravgen, Inc. v. Laboratory Corp. of America Holdings et al.*, C.A. No. 20-969 (W.D. Tex.); and *Ravgen, Inc. v. Quest Diagnostics*, No. 2:21-cv-09011(C.D. Cal.). Pet. 3; Paper 4; Paper 9.

Petitioner and Patent Owner also identify other matters involving the '720 patent before the Patent Office. Pet. 3–4; Paper 4, 1–2; Paper 9, 1. Claims of the '720 patent have been challenged in IPR2021-00791 (filed by Quest Diagnostics (“Quest”)) and IPR2021-01026 (filed by Laboratory

Corporation of America (“Labcorp”)). The ’720 patent is also involved in Ex Parte Reexamination Control Nos. 90/014,869 and 90/014,703. On October 8, 2021, we instituted trial in IPR2021-00791 and on December 14, 2021, we instituted trial in IPR2021-01026. We terminated IPR2021-00791 due to settlement (IPR2021-00791, Paper 66), and entered a final written decision on December 12, 2022, in IPR2021-01026 (finding that Labcorp had not shown that the challenged claims were unpatentable). The reexaminations have been stayed. *See* IPR2021-00791, Paper 25 (staying Ex Parte Reexamination Control No. 90/014,703); IPR2021-01026, Paper 17 (staying Ex Parte Reexamination Control No. 90/014,869).

The related ’277 patent has also been challenged in several matters before the Office: IPR2021-00788 (filed by Quest and terminated due to settlement (IPR2021-00788, Paper 71)), IPR2021-00789 and IPR2021-00790 (both filed by Quest and denied institution), IPR2021-00902 and IPR2021-01054 (both filed by Labcorp; final written decisions entered on November 1, 2022), IPR2021-01577 (filed by Streck, Inc.; currently pending), and IPR2021-01272 (filed by Illumina; final written decision entered concurrently with this Decision). The ’277 patent is also challenged in *Ex Parte* Reexamination Control No. 90/014,792, which has been stayed. *See* IPR2021-00902, Paper 24 (granting stay).

B. Asserted Grounds of Unpatentability

Petitioner asserts five grounds of unpatentability in its Petition (Pet. 4–5), which are provided in the table below:

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1, 3–8, 21, 24	102(a), (e) ⁴	Landes ⁵
22	103(a)	Landes, Marx ⁶
1, 3–6, 24	102(b)	Yen-Lieberman ⁷
7–9, 21, 25–30	103(a)	Yen-Lieberman, Gocke ⁸
22	103(a)	Yen-Lieberman, Gocke, Marx

Petitioner relies on the declarations of Dr. Brynn Levy, among other evidence. Ex. 1043; Ex. 1068. During trial, Patent Owner submitted and relies on a declaration from Dr. Brian Van Ness (Ex. 2239), among other

⁴ The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) (“AIA”), amended 35 U.S.C. §§ 102 and 103. Based on the filing date of the ’720 patent, we apply the pre-AIA versions of §§ 102 and 103.

⁵ Landes et al., International Publication No. WO 03/062441, published July 31, 2003 (Ex. 1003, “Landes”).

⁶ Marx et al., *Reducing White Blood Cells in Platelet Units*, 31(8) Transfusion 743–747 (1991) (Ex. 1029, “Marx”).

⁷ Yen-Lieberman, et al., *Diagnosis of Human Cytomegalovirus Disease*, 22(14) Clinical Microbiology Newsletter 105–109 (2000) (Ex. 1033, “Yan-Lieberman”).

⁸ Gocke et al., U.S. Patent No. 6,156,504, issued Dec. 5, 2000 (Ex. 1034, “Gocke”).

evidence.⁹ The deposition testimony of Drs. Levy (Exs. 2294, 2299) and Van Ness (Exs. 1067, 1069, and 1070¹⁰) is also of record.

C. Technology Overview and the '277 Patent

The '720 patent discloses a “rapid, non-invasive method for determining the sequence of DNA from a fetus” that can be used “for detection of chromosomal abnormalities.” Ex. 1001, 1:23–35. The '720 patent explains that a variety of invasive and non-invasive techniques are available for prenatal diagnosis, including amniocentesis, fetal blood cells in maternal blood, and maternal serum alpha-fetoprotein. *Id.* at 2:42–46. According to the patent, however, “techniques that are non-invasive are less specific, and the techniques with high specificity and high sensitivity are highly invasive.” *Id.* at 2:46–49; *see also id.* at 3:24–27 (citing an increased fetal mortality risk of about 0.5% with amniocentesis).

The '720 patent acknowledges the prior non-invasive use of fetal cells and cell-free fetal DNA (“cffDNA”), both isolated from maternal blood, for prenatal diagnosis. *Id.* at 4:59–5:44. With regard to fetal cells, the patent notes that the “presence of fetal nucleated cells in maternal blood makes it possible to use these cells for noninvasive prenatal diagnosis,” and that such “cells can be sorted and analyzed by a variety of techniques to look for

⁹ The parties submit, for example, additional testimony related to products that are alleged to practice the claimed subject matter and asserted commercial success of such products. *See, e.g.*, Exs. 2080 (Declaration of Paul K. Meyer) and 2242 (Declaration of Jeffrey Chalmers, Ph.D.).

¹⁰ Exhibit 1069 is deposition testimony taken in IPR2021-01026, and Exhibit 1070 is deposition testimony taken in IPR2021-00791.

particular DNA sequences.” *Id.* at 4:60–65. Yet the patent states that “it is still difficult” to get many fetal cells from maternal blood and “[t]here may not be enough to reliably determine anomalies of the fetal karyotype or assay for other abnormalities.” *Id.* at 5:15–21. The ’720 patent states that fetal DNA “has been detected and quantitated in maternal plasma and serum” and that “fetal DNA present in the maternal serum and plasma is comparable to the concentration of DNA obtained from fetal cell isolation protocols.” *Id.* at 5:24–34. “However,” according to the patent, “the diagnostic and clinical applications of circulating fetal DNA is limited to genes that are present in the fetus but not the mother” and that “a need still exists for a non-invasive method that can determine the sequence of fetal DNA and provide definitive diagnosis of chromosomal abnormalities in a fetus.” *Id.* at 5:38–44.

The ’720 patent describes a method that is said to increase the proportion or percentage of the cffDNA component in a sample from a pregnant female for subsequent analysis. According to the ’720 patent, the ability to detect chromosomal abnormalities has been “hindered by the low percentage of free fetal DNA” in maternal samples. Ex. 1001, 89:26–30. “Increasing the percentage of free fetal DNA would enhance the detection” of mutations, trisomy, and other genetic abnormalities. *Id.* at 89:30–35.

With the aim of increasing the percentage of cffDNA relative to circulating maternal DNA in a maternal sample, the ’720 patent describes adding an agent that inhibits cell lysis. *Id.* at 210:19–24 (Example 15) (“[T]he use of cell lysis inhibitors, cell membrane stabilizers, or cross-linking reagents can be used to increase the percentage of fetal DNA in the

maternal blood.”); *id.* at 89:35–37 (“The percent of fetal DNA in plasma obtained from a pregnant female was determined both in the absence and presence of inhibitors of cell lysis”). The ’720 patent explains that, “[w]hile lysis of both maternal and fetal cells is inhibited, the vast majority of cells [in a maternal blood sample] are maternal, and thus by reducing the lysis of maternal cells, there is a relative increase in the percentage of free fetal DNA.” *Id.* at 33:42–45. The patent identifies numerous agents as cell lysis inhibitors, cell membrane stabilizers, or cross-linking reagents. *See, e.g., id.* at 32:65–33:28 (listing, for example, formaldehyde, formalin, cleavable crosslinkers, cholesterol, vitamin E and its derivatives, and glucose); *see also id.* at 6:49–65; 11:43–12:5; 216:14–218:55 (Table XXIII (“A representative list of cross-linkers that can be used to impede maternal cell lysis”)).

The ’720 patent provides results on the use of formalin (i.e., formaldehyde in aqueous solution) as the lysis-inhibiting agent. *Id.* at 89:25–92:26 (Example 4), 210:16–218:55 (Example 15).¹¹ In Example 4, the patent describes collecting a 5 ml blood sample from a pregnant subject, separating the sample into two tubes (each containing EDTA¹²), and adding

¹¹ The parties have used formaldehyde and formalin interchangeably in this proceeding, noting simply that formalin is formaldehyde in water. Pet. 2 (“formaldehyde (referred to as formalin when in solution)”), 10 (“This petition also discusses ‘formalin,’ which is a liquid that contains formaldehyde and water”); Ex. 2294, 108:4–10 (testimony of Dr. Levy that he is using the terms formalin and formaldehyde “interchangeably”).

¹² The ’720 patent states that EDTA is a “magnesium chelator.” Ex. 1001, 32:61–64.

formaldehyde (25µl/ml) to one of the tubes. *Id.* at 89:40–46. The samples were centrifuged and 800 µl of each maternal plasma sample was then used for DNA purification and further processing to determine the relative amount of cffDNA in the samples. *Id.* at 89:47–92:26. According to the '720 patent, “the percentage of fetal DNA present in the sample that was treated with only EDTA was 1.56%” and the “percentage of fetal DNA present in the sample treated with formalin and EDTA was 25%.” *Id.* at 91:34–41. The patent reports that the percent of total cffDNA in eighteen blood samples with and without formalin was then calculated, with the results (mean percentage cffDNA) provided in Table V. *Id.* at 91:45–92:10 (reporting, *inter alia*, a mean percentage fetal DNA of 19.47% with formalin versus 7.71% without formalin); *see also id.* at 210:16–215:65 (Example 15 (reporting further results on formalin use in Table XXI)).

D. Challenged Claims

The '720 patent includes thirty claims. Of those, claims 1, 3–9, 21, 22, and 24–30 are challenged here. Pet. 4. Claim 1, the only independent claim, is illustrative of the claims challenged in this Petition and reads as follows:

1. A method for detecting a free nucleic acid, wherein said method comprises: (a) isolating free nucleic acid from a non-cellular fraction of a sample, wherein said sample comprises an agent that impedes cell lysis, if cells are present, and wherein said agent is selected from the group consisting of membrane stabilizer, cross-linker, and cell lysis inhibitor; and (b) detecting the presence or absence of the free nucleic acid.

Ex. 1001, 535:15–21.

E. Prosecution History

The application that issued as the '720 patent was filed on August 26, 2005. Ex. 1001, codes (21), (22). The Examiner initially rejected pending claim 1 as anticipated by Kiessling. Ex. 1012, 1145. The Examiner also rejected claim 1 as obvious over the combination of Kiessling and Anker. *Id.* at 1149–50. With respect to what the Examiner described as the “key limitation i.e., wherein an agent that impedes cell lysis is added to the sample,” the Examiner found that Kiessling disclosed adding formaldehyde, paraformaldehyde, or glutaraldehyde to the sample to impede cell lysis. *Id.* at 1145. The Examiner further found that Kiessling taught that “the addition [of] one of these fixatives fixes (i.e. impedes the cell lysis of) the leukocytes present in the sample (i.e. human blood) and disinfects the sample.” *Id.*

In response, Patent Owner amended claim 1 to recite that the nucleic acid detected and isolated was “free” nucleic acid and to specify that the agent that impedes cell lysis was “selected from the group consisting of membrane stabilizer, cross-linker, and cell lysis inhibitor.” *Id.* at 1180. Patent Owner then argued that amended claim 1 was distinguishable over Kiessling because Kiessling analyzed DNA from within leukocytes while the amended claims focus on free nucleic acid. *Id.* at 1189–90 (“The methodology disclosed in Kiessling ‘fixes’ leukocytes, and subsequently DNA is isolated from the fixed leukocytes. Conversely, the methods of [the pending claims] are directed to isolating free nucleic acid. Nowhere does Kiessling teach or suggest a method for detecting free nucleic acid comprising *inter alia* isolating free nucleic acid from a sample.”) (internal

citation omitted). Patent Owner also argued that the claimed method solved long-felt needs and achieved unexpected results. *Id.* at 1191–1192.

Initially the Examiner found Patent Owner’s amendment and arguments persuasive, withdrawing the rejections based on Kiessling and Anker and explaining that “[t]he most persuasive argument(s) related to the applicant[’s] assertion that there was a long felt need and the evidence of an unexpected result.” *Id.* at 2100. Shortly thereafter, however, the Examiner reinstated the anticipation rejection over Kiessling. *Id.* at 2124. In the same office action, the Examiner also made several obviousness rejections of certain of the dependent claims over Kiessling alone and in combination with various other prior art references. *Id.* at 2128–2132. In doing so, the Examiner explained that “absent an unexpected result [the claimed method] would have been *prima facie* obvious.” *Id.* at 2128, 2130, 2131.

In response, Patent Owner argued that the Examiner misunderstood Kiessling, again arguing that the claimed method was distinguishable over Kiessling because “[t]he Kiessling methods do not involve the isolation of free nucleic acid from samples, and therefore, do not teach methods for detection of free nucleic acid.” *Id.* at 2148–49. Patent Owner also repeated its argument that the claimed method solved a long felt need and demonstrated unexpected results. *Id.* at 2150.

The Examiner responded by maintaining the anticipation rejection over Kiessling and adding obviousness rejections over several different combinations of art. *Id.* at 2168–74. The Examiner added an obviousness rejection over the combination of Amicucci or Saito in view of Kiessling, finding that “both Amicucci et al. and Saito et al. teach a method of

detecting free nucleic acid which comprises all of the limitations of Claim 1, except these authors do not teach that their maternal samples comprise[] an agent that impedes cell lysis.” *Id.* at 2168. The Examiner nonetheless found claim 1 obvious because Kiessling taught the use of a cell lysis inhibitor. *Id.*

Patent Owner responded by amending claim 1 to recite that the free nucleic acid is isolated from “a non-cellular fraction” of the sample. *Id.* at 2238. Appellant also argued that the claimed method was distinguishable over Kiessling because “Kiessling ‘fixes’ leukocytes, and subsequently DNA is isolated from the fixed leukocytes,” i.e., “isolation of DNA from a cellular fraction.” *Id.* at 2245. Thus, according to Patent Owner, Kiessling did not teach or suggest detecting free nucleic acid from a non-cellular fraction of a sample. *Id.* Patent Owner also repeated its argument that the claimed method solved a long felt need and demonstrated unexpected results. *Id.* at 2247–48.

The Examiner subsequently stated that the pending claims were “deemed to be allowable over the prior art of record in light of the applicant’s amendment filed 02 JUN 09 and the persuasive argument(s) therein.” *Id.* at 2271. After the filing of a terminal disclaimer to address a non-statutory obviousness-type double patenting rejection (*id.* at 2280), the Examiner issued a notice of allowance (*id.* at 2286–88).

II. ANALYSIS

A. Principles of Law

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is

unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim”))).

To show anticipation under 35 U.S.C. § 102, each and every claim element, arranged as in the claim, must be disclosed in a single piece of prior art. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359 (Fed. Cir. 2008). “An element may be inherently disclosed only if it is necessarily present, not merely probably or possibly present, in the prior art.” *Guangdong Alison Hi-Tech Co. v. Int’l. Trade Comm’n*, 936 F.3d 1353, 1364 (Fed. Cir. 2019). “[A] limitation or the entire invention is inherent and in the public domain if it is the natural result flowing from the explicit disclosure of the prior art.” *Schering Corp. v. Geneva Pharm.*, 339 F.3d 1373, 1379 (Fed. Cir. 2003) (internal quotation and citation omitted).

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which that subject matter pertains. *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) secondary considerations of nonobviousness when presented. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). When evaluating a combination of teachings, we must also

“determine whether there was an apparent reason to combine the known elements in the fashion claimed by the patent at issue.” *KSR*, 550 U.S. at 418 (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)). Whether a combination of elements produces a predictable result weighs in the ultimate determination of obviousness. *Id.* at 416–417.

B. Person of Ordinary Skill in the Art (“POSA”)

In determining the level of skill in the art, we consider the problems encountered in the art, the art’s solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Petitioner proposes the following POSA definition:

[A POSA] would have had “a M.D. and/or Ph.D. in a related area such as genetics, biochemistry, molecular biology, cell biology, or microbiology and at least one to two years of work in one of those related areas . . . [or] a Bachelor’s degree in one of the foregoing areas and at least three to four years of work in that area.” . . . Based on this asserted level of skill, the POSA at the relevant time would have had academic or industry experience collecting and analyzing cell-free nucleic acid, such as DNA, from blood samples.

Pet. 12 (quoting Ex. 1020 ¶ 15); *see also* Ex. 1043 ¶¶ 13–15.

We applied this definition at the institution stage. Inst. Dec. 36–37. Patent Owner did not expressly contest this definition during trial, noting only that “Dr. Van Ness meets [this] definition” and “confirms [that] his opinions are the same under either party’s definition.” PO Resp. 6 (citing Ex. 2239 ¶¶ 20–25). To the extent there are minor differences in the above

definition and the alternative definition offered by Dr. Van Ness (Ex. 2239 ¶ 23), Dr. Van Ness’s proposal is too broad for at least the reason that it does not require the POSA have experience collecting and analyzing cell-free DNA from blood samples—subject matter relevant to the claims and applied art. Accordingly, we continue to apply Petitioner’s proposed definition here.¹³ In any event, the parties’ technical declarants meet or exceed either definition and there is no indication that their opinions would change depending on which definition was applied, and no argument from the parties to the contrary. Ex. 1056 ¶¶ 5–10 (overview of qualifications and experience); Ex. 2239 ¶¶ 4–7 (same).

C. Claim Construction

We interpret a claim “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b). Under this standard, we construe the claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.*

A patent’s specification may provide a special definition to a claim term, or the intrinsic evidence may reveal an inventor’s intentional disclaimer or disavowal of claim scope. *Phillips v. AWH Corp.*, 415 F.3d.

¹³ We adopted a somewhat differently phrased definition for the POSA in the Labcorp cases based on the parties’ proposals in those matters. *See, e.g.*, IPR2021-01026, Paper 51 at 9–11. On this record, we do not see that such differences are material or would alter the outcome of this Decision.

1303, 1316 (Fed. Cir. 2005). If the inventor acts as their own lexicographer to give a claim term a special definition, that definition must be set forth in the specification with reasonable clarity, deliberateness, and precision. *Renishaw PLC v. Marposs Societa' per Azioni*, 158 F.3d 1243, 1249 (Fed. Cir. 1998). A disavowal, if any, can be made by language in the specification or the prosecution history. *Poly-America, L.P. v. API Indus., Inc.*, 839 F.3d 1131, 1136 (Fed. Cir. 2016). “To disavow claim scope, the specification must contain ‘expressions of manifest exclusion or restriction, representing a clear disavowal of claim scope.’” *Retractable Techs., Inc. v. Becton, Dickinson & Co.*, 653 F.3d 1296, 1306 (Fed. Cir. 2011) (quoting *Epistar Corp. v. Int’l Trade Comm’n*, 566 F.3d 1321, 1335 (Fed. Cir. 2009)). “[T]he standard for disavowal is exacting, requiring clear and unequivocal evidence that the claimed invention includes or does not include a particular feature.” *Poly-America*, 839 F.3d at 1136.

Petitioner does not identify any claim terms as needing express construction. Pet. 21–22 (“[T]he Board need not construe any terms of the challenged claims to resolve the underlying controversy.”). Patent Owner argues that the claims should be construed to exclude alleged “anticoagulant chelators,” including EDTA and ACD, as satisfying the “agent” term. PO Resp. at 7. We discuss in more detail below.

Claim 1 recites, among other limitations, that the sample comprises “an agent that impedes cell lysis, if cells are present.” *See supra* Section

I(D).¹⁴ Patent Owner, citing a claim construction adopted by the district court in Patent Owner’s litigation with third-party Natera, argues that this “agent” limitation should be construed the same way by the Board. PO Resp. 6–7 (citing Ex. 2039, 1). The district court interpreted the “agent” limitation to mean:

[A] substance that inhibits the rupture of cells that is selected from the group consisting of membrane stabilizer, cross-linker and cell lysis inhibitor, and *does not include chelators used as anticoagulants* nor endogenous substances.

Id.; Ex. 2039, 1 (Supplemental Claim Construction Order) (emphasis added).¹⁵ According to Patent Owner, “[t]he specification, prosecution histories [of the ’277 and ’720 patents], and extrinsic evidence consistently and unambiguously support” this construction. PO Resp. 7. Because “chelators used as anticoagulants” encompasses both EDTA and ACD, Patent Owner argues, the “Board should thus construe the agent term to exclude anticoagulant chelators, including EDTA and ACD.” *Id.*

¹⁴ The related ’277 patent includes claims that require “an agent that inhibits lysis of cells, if cells are present.” See IPR2021-01272, Ex. 1001, 473:2–3. We use the terms “inhibits” and “impedes” interchangeably, as the parties have done in these proceedings.

¹⁵ Pursuant to our rules, we take this claim construction into account in construing the claims in this proceeding. 37 C.F.R. § 42.100. Although this construction also excludes “endogenous substances,” we need not further address that language because no challenge in this case cites endogenous substances as being the claimed “agent.” *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Ltd.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (“[W]e need only construe terms that are in controversy, and only to the extent necessary to resolve the controversy.”) (internal quotation marks omitted).

Patent Owner cites intrinsic evidence in support of its interpretation. *Id.* at 7–14. For example, Patent Owner contends the Specification distinguishes anticoagulant chelators from the claimed agent, including in a working example (Example 4) where a blood sample placed in a tube with *only EDTA* was treated as a control and compared with a sample purporting to illustrate the invention, which included *EDTA in combination with formalin*. *Id.* at 7–8 (citing Ex. 1001, 89:35–37, 89:46–47, 90:16–20). This example, the patent discloses, shows processing of blood samples “both in the absence and presence of cell lysis inhibitors,” which Patent Owner contends should be reasonably read as indicating that an agent that inhibits lysis of cells is “absent” in the EDTA-only sample. *Id.* (emphasis omitted).¹⁶

Furthermore, Patent Owner cites portions of the prosecution history where the applicant argued that disclosures about EDTA in the applied art did not meet the “agent” term. *Id.* at 8–13 (citing, applicant’s argument during prosecution of the ’277 patent that the Office was “simply incorrect” in finding that EDTA in a prior art reference relied upon by the Examiner was an agent that inhibits lysis of cells; “Rather, EDTA is a well-known chelator of calcium and magnesium . . . routinely added to blood . . . as an anticoagulant due to its ability to chelate calcium.”) (emphasis omitted).

¹⁶ The ’720 patent describes EDTA as a “magnesium chelator” and discusses EDTA separately from the listing of agents that inhibit cell lysis. *Compare* Ex. 1001, 32:61–63 (identifying EDTA), *with id.* at 33:12–33:28 (listing formaldehyde and glucose as, respectively, a cell lysis inhibitor and membrane stabilizer).

According to Patent Owner, similar acknowledgments were made during prosecution of the '720 patent about art that disclosed processing samples in ACD tubes. *Id.* at 11–12 (citing Ex. 1012, 1156 (Adams rejection), 2170 (Holodniy rejection); Ex. 2075, 2 (Holodniy disclosing ACD); Ex. 2023, 20:52–54 (Adams disclosing ACD)). Patent Owner contends that, during reexamination of the '720 patent, it again disavowed any claim scope that would encompass ACD. PO Resp. 13 n.1 (citing Ex. 2238, 30–34, and *Aylus Networks, Inc. v. Apple Inc.*, 856 F.3d 1353, 1361 (Fed. Cir. 2017)).¹⁷

Finally, Patent Owner cites extrinsic evidence, including several technical publications as showing that EDTA and ACD are routinely

¹⁷ Although a patentee's statements before the Office may evidence a disclaimer, Patent Owner's statements in the cited reexamination do not weigh heavy here. There is no record here that the Examiner has acted on Patent Owner's statements of alleged disavowal in the reexamination. In *Aylus*, by contrast, the district court relied on patentee's earlier statements made in its preliminary response that contributed to the Board's *denial of inter partes review* for the relevant claims. *Aylus*, 856 F.3d at 1362–63. In that light, the patentee could not advance a narrow claim interpretation before the Office that secured IPR denial and later maintain a broader (and inconsistent) interpretation before the courts, which the courts in *Aylus* viewed as a prosecution disclaimer. Here, however, the reexamination of the '720 patent had just begun and was stayed so that the related IPRs could be completed first. We do not treat Patent Owner's argument *in the present IPR* about the claims not covering ACD or its chemical components as a "disclaimer," and we decline to elevate similar argument made in the stayed reexam to such status. *Cf. Cupp Computing AS v. Trend Micro Inc.*, 53 F.4th 1376, (Fed. Cir. 2022) (holding "a disclaimer is not binding on the PTO in the very proceeding in which it is made, just as a disclaimer in a district court proceeding would not bind the district court in that proceeding").

characterized as anticoagulants for blood-collection tubes, not as cell-lysis inhibitors, fixatives, or preservatives. PO Resp. 13–14. Patent Owner cites, for example, the FDA labeling for ACD classifying it as an anticoagulant and noting that ACD “acts as an anticoagulant by the action of the citrate ion chelating free ionized calcium.” *Id.* at 13 (quoting Ex. 2071, 1). Patent Owner also cites, for example, a published patent application from tube-maker Streck, Inc., that lists EDTA and ACD among “anticoagulants,” while listing preservatives and polysaccharides separately. *Id.* at 14–15 (citing Ex. 2073 ¶¶ 14 (listing formaldehyde as a preservative), 89 (listing anticoagulants), 90 (listing polysaccharides)).

The question that Patent Owner’s proposed claim construction presents is whether a claimed agent that inhibits cell lysis may be a chelator used as an anticoagulant. The Specification and prosecution history make clear that at least one anticoagulant chelator—EDTA—is excluded from satisfying the claimed agent as recited in claim 1. As discussed above, the Specification describes an example where a sample processed in a tube with only EDTA is described as processing a sample in the “absence” of an agent that inhibits cell lysis. Ex. 1001, 89:35–57 (describing, in contrast, the EDTA + formalin sample as showing an example in the “presence of inhibitors of cell lysis”). If EDTA alone could be read as meeting the “agent” limitation, that example would make no sense. Petitioner provides no persuasive argument otherwise.

The prosecution history buttresses the exclusion of the anticoagulant chelator EDTA from being the claimed “agent.” For example, during prosecution of the related ’277 patent, the Examiner at one point expressly

identified the EDTA in the tubes of a prior art reference as satisfying the agent term, prompting applicant's argument that the Office was "incorrect" in doing so, and further distinguishing EDTA as merely "a well-known chelator of calcium and magnesium" that is added to blood "as an anticoagulant due to its ability to chelate calcium." Ex. 2064, 1196 (arguing "EDTA is clearly referred to as a chelator in Applicant's specification, not as a cell lysis inhibitor"); *supra* Section I(E); Ex. 1001, 32:61–64 (listing EDTA and EGTA as chelators). This, in turn, convinced the Examiner to withdraw the finding that EDTA was an "agent" as claimed and move on to art that disclosed formaldehyde as meeting that claim term. Ex. 2064, 1024 (Examiner interview summary noting that "applicant pointed out [that] EDTA could not be defined as a cell lysis inhibitor but rather was simply an anticoagulant"). This colloquy during prosecution evidences a clear and unmistakable disavowal of EDTA as being the claimed "agent."

Having determined that Patent Owner disavowed EDTA as the claimed "agent," we next consider whether the disavowal extends more broadly to other anticoagulant chelators and, specifically, to ACD. On this record, we find that it does as explained below. In distinguishing EDTA from being the "agent," Patent Owner relied not on properties that are unique to EDTA, but on the fact that it was a "well-known chelator" used as an anticoagulant in this technical field. Ex. 2064, 1196. Given Patent Owner's clear intent to exclude EDTA from the scope of the claimed "agent" on the basis that it was simply an anticoagulant chelator, we find that a POSA would also have understood other anticoagulant chelators as subject to Patent Owner's disavowal.

Consistent with this understanding, during prosecution of the '720 patent, the Examiner found that Holodniy (and at least one other reference) did not disclose the “agent” term—notwithstanding that such references disclosed processing samples in ACD tubes. Ex. 2041, 2170 (finding that Holodniy “teach[es] a method of detecting free nucleic acid which comprises all of the limitations of Claim 1, except these authors do not teach that their samples comprise an agent that impedes cell lysis”); *see also id.* at 1156 (finding Adams “do[es] not teach adding an agent that impedes cell lysis to the sample”). Patent Owner confirmed the Examiner’s finding on this point. *Id.* at 2254 (asserting that Holodniy fails to teach or suggest the claimed method “wherein an agent that impedes cell lysis has been added”). And Patent Owner argues persuasively, without dispute from Petitioner, that ACD is the only addition to the collection tubes of Holodniy and Adams besides the sample itself. PO Resp. 12; PO Sur-reply 2–3; Ex. 2075, 2; Ex. 2023, 20:52–54.¹⁸ This lends further support that neither the Examiner nor Patent Owner considered ACD as an anticoagulant chelator to be an “agent” as claimed.

For the reasons above, we construe the phrase “an agent that impedes cell lysis” to exclude chelators used as anticoagulants. Although the “agent” cannot itself be an anticoagulant chelator, the sample, as a whole, in the

¹⁸ It appears to be the case, however, that ACD was never explicitly highlighted by the Examiner or applicant during prosecution in relation to the agent term or otherwise. As we discuss in more detail below, ACD is not a single chemical compound; instead it includes multiple compounds in solution, and some of them are not chelators. Ex. 1068 ¶ 3; Ex. 2071, 1; Ex. 2074, 4; Ex. 1067, 51:14–17; Ex. 1069, 114:17–20; Ex. 1067, 56:11–16.

claimed methods may still *include* chelators used as anticoagulants. The claims use the phrase “said sample *comprises*” indicating that the claims are open ended, and that the sample may include other unrecited components. *Exergen Corp. v. Wal-Mart Stores, Inc.*, 575 F.3d 1312, 1319 (Fed. Cir. 2009) (“The claim uses the term ‘comprising,’ which is well understood in patent law to mean ‘including but not limited to.’”). Indeed, as explained above, the Specification includes examples that purport to represent the invention where maternal blood samples are processed in tubes that combine EDTA (an anticoagulant chelator) with formalin/formaldehyde (the cell-lysis inhibitor), which samples are contrasted with a control where the maternal blood is processed in EDTA-only tubes.¹⁹

D. Overview of the Asserted Prior Art

1. Landes (Exhibit 1003)

Landes is an international patent application that published July 31, 2003. Ex. 1003, code (43). Landes claims priority to a U.S. provisional patent application (No. 60/349,877) that was filed January 18, 2002 (hereafter “the Landes Provisional” (Ex. 1006)).

Landes relates generally to non-invasive methods used to distinguish

¹⁹ Because we find that the intrinsic evidence is sufficiently clear, we need not rely on extrinsic evidence for purposes of construing the claims. That said, based on the present record, consistent with Patent Owner’s argument (PO Resp. 13–15), ACD is commonly characterized as an anticoagulant (not a preservative) in many external publications. But that is not always the case, as is plain from other extrinsic evidence of record. *See, e.g.*, Ex. 1062, 613 (describing ACD as a “preservative” of blood or blood cells).

circulating fetal DNA from maternal DNA in samples from maternal blood, and thereby to detect fetal aneuploidies and alleles. Ex. 1003, code (57). Landes teaches, for example, that “maternal serum or plasma may be a relatively rich source of fetal DNA based on PCR determinations,” that “fetal DNA can be consistently detected in maternal serum as early as 7 weeks,” and that such DNA increases during gestation. *Id.* at 2:19–25 (disclosing that “the lowest fetal DNA concentration in plasma as measured by PCR was greater than 20 fetal cell equivalents per mL of maternal blood with some instances where fetal DNA constituted as much as 5% of the total DNA in plasma”).

Landes discloses a plasma separation and fetal DNA isolation protocol where maternal blood is first collected into ACDA tubes supplied from Becton, Dickinson. *Id.* at 10:26–12:6. Landes teaches that, following centrifugation and plasma removal, DNA can be isolated from the plasma sample using commercially available kits. *Id.* Landes teaches that the DNA may then be treated with bisulfite to distinguish fetal and maternal DNA and the nucleotide sequence of interest may be subsequently amplified (e.g. via fetal-specific PCR) and analyzed. *Id.* at 12:9–15:3. More specifically, Landes teaches that isolated DNA may be treated “with a reagent which differentially modifies methylated and non-methylated DNA, e.g., bisulfite,” which recognizes that fetal DNA is generally “hypomethylated relative to adult DNA.” *Id.* at 8:22–28 (explaining that “[o]ne means of generating

fetal-specific PCR products is to identify loci that are unmethylated in fetal DNA and methylated in adult/maternal DNA.”).

2. Marx (Exhibit 1029)

Marx is an article that published in 1991. Ex. 1029, 743. Marx relates to techniques that aim to prepare platelet rich plasma samples with reduced white blood cell contamination. *Id.* Marx teaches, for example, “an effective method of significantly decreasing the WBC content” involves “reduced centrifugation time and with the braking programs disengaged.” *Id.*; *see also id.* at 746 (“[N]o brakes gave much more reasonable yields . . . as well as low WBC contamination.”).

3. Yen-Lieberman (Exhibit 1033)

Yen-Lieberman is an article entitled “Diagnosis of Human Cytomegalovirus Disease,” that appears in the Clinical Microbiology Newsletter. Ex. 1033. The article is dated July 15, 2000. Yen-Lieberman is a review of the “methodological aspects of newer and current commercially available tests” for detecting human cytomegalovirus (“HCMV”). *Id.* at 108. One of the tests discussed is the AMPLICOR CMV test, which “uses PCR nucleic acid amplification for qualitative detection of HCMV DNA in plasma separated from anticoagulated whole blood (EDTA or ACD anticoagulants).” *Id.* at 107. Yen-Lieberman teaches that “the detection of circulating HCMV DNA in plasma or whole blood predicts the subsequent development of HCMV disease.” *Id.* at 108 (internal citations omitted). “Additionally, quantitation of the amount of HCMV DNA in blood is a useful clinical tool to monitor patients on therapy.” *Id.*

4. Gocke (Exhibit 1034)

Gocke is an issued U.S. Patent, entitled “Detection of Extracellular Tumor-Associated Nucleic Acid in Blood Plasma or Serum Using Nucleic Acid Amplification Assays.” Ex. 1034, code (54). It issued on December 5, 2000. *Id.* at code (45). Gocke discloses “detection of specific extracellular nucleic acid in plasma or serum fractions of human or animal blood.” *Id.* at Abstract. Gocke states that the methods disclosed therein can be used to detect “[n]ucleic acid from any biological source, including but not limited to . . . viral . . . nucleic acid.” *Id.* at 7:57–60.

E. Ground 1: Anticipation by Landes

Petitioner asserts that claims 1, 3–8, 21, and 24 are unpatentable as anticipated by Landes. Pet. 22–32. Petitioner contends Landes is prior art under §§ 102(a) and/or (e). *Id.* at 13.²⁰ With respect to the “agent that

²⁰ Before institution, Patent Owner contested the sufficiency of Petitioner’s showing on whether Landes was entitled to the priority benefit of the filing date of the Landes provisional application. We disagreed that Petitioner’s showing was insufficient. Inst. Dec. 43–47. Patent Owner did not raise any issue during trial about Landes’s prior art status and, thus, has forfeited any argument that Landes is not prior art. On this record, we determine that Landes is § 102(a) prior art based on its July 31, 2003, publication date. On this record, we do not presume the claims of the ’720 patent are entitled to any earlier priority date of other applications listed in its ancestral chain. *PowerOasis, Inc. v. T-Mobile USA, Inc.*, 522 F.3d 1299, 1305 (Fed. Cir. 2008) (explaining that “there is simply no reason to presume that claims in a [CIP] application are entitled to the effective filing date of an earlier filed application”). That is especially true with the patent’s complex priority chain, involving several CIPs leading back to a provisional application filed

impedes cell lysis, if cells are present” limitation, Petitioner points to Landes’s disclosure of a method for collecting maternal blood in tubes that contain ACD—and, more specifically, that such tubes will necessarily expose the blood to dextrose (i.e., glucose).²¹ *Id.* at 24–28 (citing, for example, Ex. 1003, 10:28–29 (disclosing use of an “ACDA blood collection tube”); Ex. 1007, 4:30–39 (listing components of Becton Dickinson’s ACD-A formulation, including sodium citrate, citric acid, and 24.5 gm of dextrose•H₂O)). According to Petitioner, “[t]he ’720 patent identifies glucose as ‘an agent that impedes cell lysis’ according to the alleged invention.” *Id.* at 24–25 (citing Ex. 1001, 33:12–28 (“In another embodiment, an agent that stabilizes cell membranes may be added to the maternal blood samples to *reduce maternal cell lysis* including . . . *glucose*.”) (Petitioner’s emphasis)).

Patent Owner raises multiple counterarguments. PO Resp. 16–36; PO Sur-Reply 2–12. First, essentially repeating its claim construction analysis, Patent Owner argues that Landes uses ACD only as an anticoagulant without disclosing any specific functionality of ACD’s dextrose component, and that

on May 8, 2002. Ex. 1001, code (60), (63). Neither party has made any showing in this proceeding that the challenged claims are entitled to such earlier priority dates. Moreover, even if it the challenged claims could claim priority to the filing dates in 2002, the Landes provisional (filed Jan. 18, 2002; Ex. 1006) predates those applications and Landes would be § 102(e) prior art based on the reasoning in the Institution Decision, which we adopt here. Inst. Dec. 43–47.

²¹ The parties did not argue any material distinction between “dextrose” and “glucose” in this proceeding and, like the parties, we use the terms interchangeably herein.

Landes's ACD should be excluded from satisfying the "agent" term. PO Resp. 17–21; *see also id.* at 26 ("[T]he sole role of ACD in Landes was to prevent coagulation so that plasma could be collected and frozen until DNA isolation"). Patent Owner contends that the public has been on notice that anticoagulant chelators are not the claimed "agent" and, accordingly, that "Landes's disclosure of ACD (and its dextrose component) fails to meet the 'agent' limitation." *Id.* at 20–21. Second, even if the dextrose of ACD is not excluded from the scope of the "agent" term as a matter of claim interpretation, Patent Owner argues that Petitioner fails to provide evidence sufficient to show that dextrose in Landes's ACD tubes and method necessarily "inhibits lysis of cells" as claimed. *Id.* at 21–22; 29–36 (arguing, *inter alia*, that a "mere possibility" of ACD's dextrose component inhibiting lysis in Landes's method is not enough); PO Sur-reply 10–12 (arguing that Petitioner's contentions about dextrose regulating "isotonicity" or providing a "nutrient source" for cells are belated and otherwise substantively flawed).²²

There is no dispute on this record that ACD includes dextrose, and that dextrose is a species of glucose. Ex. 1043 ¶ 24 (Dr. Levy testimony that ACD includes dextrose, the dextrorotatory form of glucose); Ex. 1068 ¶ 3 (Dr. Levy listing components of ACD); Ex. 2074, 4; Ex. 1067 54:2–6 (Dr.

²² Patent Owner also argues that Landes does not disclose the claim step of "detecting the presence or absence of the free nucleic acid" because Landes's method modifies portions of the cffDNA, changing certain nucleotides (i.e., C to T) in those DNA sequences. PO Resp. 36–39. We do not reach this argument because, as explained herein, Petitioner's challenge to the claims fails for other reasons.

Van Ness agreeing that, “[b]y definition, an ACD tube to a person of skill in the art would contain glucose”). Nor is there any dispute that the ’720 patent identifies glucose as among several cell-stabilizing agents that may be added to a sample to inhibit maternal cell lysis. Ex. 1001, 6:49–65. Rather, the dispute here centers on two questions: 1) whether the exclusion of anticoagulant chelators from meeting the claimed “agent” limitation operates to exclude dextrose introduced to a sample through ACD, and 2) whether Petitioner has established, by a preponderance of the evidence, that the dextrose in ACD actually impedes lysis of cells in Landes’s method. We address each of these questions in turn below.

1. Whether exclusion of anticoagulant chelators as the claimed “agent” also excludes glucose introduced to a blood sample in an ACD tube?

As discussed above (Section II(C)), the claims exclude anticoagulant chelators from being the claimed “agent that impedes cell lysis, if cells are present.” We now consider whether that exclusion extends to the individual components of ACD—and dextrose, in particular. First, we provide a brief overview of ACD.

ACD is a solution comprising three main components: (1) citric acid (2) sodium citrate and (3) dextrose. Ex. 1068 ¶ 3 (testimony of Dr. Levy); Ex. 2074 (FDA prescribing information for ACD A), 1 (indicating that the “[g]eneric name” for ACD A is of “citric acid monohydrate, dextrose monohydrate, and trisodium citrate dihydrate,” and identifying the “[d]osage form” as “injection, solution”), 4 (describing mechanism of action and identifying the function of each of the three components of ACD). As Dr.

Levy explains, “it is the citrate ion chelating free ionized calcium” that acts as an anticoagulant. Ex. 1068 ¶ 3; Ex. 2074, 4 (noting ACD Solution A “acts as an extracorporeal anticoagulant by binding the free calcium in the blood,” that “[c]alcium is a necessary co-factor to several steps in the clotting cascade,” and that “[s]odium [c]itrate anticoagulates”). The citric acid provides for “pH regulation” and dextrose is for “isotonicity”—influencing movement of water across the cellular membrane. Ex. 1068 ¶ 3 (Dr. Levy stating, “citric acid, for pH regulation” and “glucose (*i.e.*, dextrose) for isotonicity”); Ex. 1067, 59:5–20 (Dr. Van Ness testifying to his understanding of “isotonicity”); Ex. 2074, 4 (stating “[c]itric acid for pH regulation” and “[d]extrose for isotonicity”). Dr. Levy testifies that “the glucose of ACD is not the anticoagulant chelator.” Ex. 1068 ¶ 3; *see also* Ex. 1067, 56:11–16 (Dr. Van Ness testifying he has no support for glucose of ACD itself acting as an anticoagulant chelator); *id.* at 58:5–15 (Dr. Van Ness testifying he has no basis to disagree with the description of ACD and its mechanism of action in the FDA prescribing information (Ex. 2074)). In short, ACD includes three main components in aqueous solution—a chelator (sodium citrate), a pH buffer (citric acid), and glucose (for isotonicity).

The ’720 expressly identifies glucose as an agent that impedes cell lysis. *See e.g.*, Ex. 1001, 6:49–56 (“An agent that stabilizes cell membranes may be added to the maternal blood sample to reduce maternal cell lysis[,], including . . . glucose”); *see also, id.* at 11:57–63; 33:12–20; 214: 4–11; 219:55–62; 254:1–8; 260:29–37; 264:14–21; 269:26–34; 274:36–43; 279:10–17; 285:1–8; 290:44–51. This presents the question whether glucose is nonetheless precluded from satisfying the claimed “agent” limitation

because it was introduced into the blood sample in a solution that also includes a chelator (sodium citrate) and a pH buffer (citric acid). We find that glucose is not so precluded. As discussed above, the claims are open-ended and may cover combinations that include an anticoagulant chelator (e.g., the compound EDTA) along with a lysis-inhibiting agent, such as formaldehyde.

The claims place no apparent limitation on how the “agent” comes to be in the sample—no such limitation was argued in this case. The claims simply require that the “sample comprises” an “agent that impedes cell lysis, if cells are present.” It would seem incongruous that a blood sample to which is added a substance expressly called out as an “agent” in the specification of the ’720 patent would be excluded from meeting the “agent” limitation simply because that substance was introduced to the sample as one component of a solution also comprising an anticoagulant/chelator. This is particularly true where there is no indication on this record that glucose (in ACD or otherwise) plays any specific role in anticoagulation or chelation. Ex. 1068 ¶ 3 (Dr. Levy testimony identifying the functions of ACD’s components); Ex. 1069, 114:17–20 (Van Ness testifying he would not classify glucose as a chelator).

For Patent Owner’s disavowal of claim scope to extend to an individual chemical component of the ACD solution that itself plays no role in anticoagulation or chelation, Patent Owner disavowal needed to make that unambiguously clear. *Poly-America*, 839 F.3d at 1136. As discussed above, the sodium citrate component of ACD is that solution’s anticoagulant

chelator—glucose, which provides a different function, is not.²³

For the reasons discussed above, Patent Owner excluded anticoagulant chelators from being the claimed “agent.” *See supra* Section II(C). It is not, however, clear that the exclusion extends to all of the individual components of a multi-part “anticoagulant” solution—at least one of which is specifically identified in the ’720 patent as being among the agents that can inhibit cell lysis. As discussed already, during prosecution the Examiner considered at least two prior art references, Holodniy and Adams, each of which collected blood in ACD tubes, and found that they lacked a teaching of the claimed agent. But there is no indication that in doing so, the Examiner considered the position urged here by Petitioner, i.e., that the glucose sub-component within ACD meets the “agent” limitation. Pet. 26 (Peticioner focusing on ACD’s glucose, and the listing of glucose in the patent); Pet. Reply 1–2 (distinguishing ACD’s citrate component as the

²³ Patent Owner cites a docket entry in *Ravgen v. Laboratory Corporation of America Holding*, 6:20-cv-00969, ECF No. 199 (W.D. Tex. Sep. 13, 2022) as the district court’s order granting summary judgment of no anticipation by Landes. PO Sur-Reply 4; Ex. 2306, 1 (“[A]cid citrate dextrose (ACD) in . . . Landes . . . do[es] not disclose the claimed ‘agent’ . . . as a matter of law.”). Beyond noting that the motion was “GRANTED,” the order includes one sentence on this point (quoted in relevant part in the parenthetical above) without further explanation of the court’s reasoning. Petitioner argues that it was not a party to that litigation, it does not have access to all the briefing and evidence underlying the order, and the court applies a higher standard of proof than is required in this IPR. Pet. Sur-Sur-Reply 1–2. For these reasons, the court’s order is of limited persuasive value and plays no role in our analysis.

anticoagulant chelator separate from glucose and its functionality). We conclude that Patent Owner’s disavowal does not unambiguously extend to glucose as relevant to Petitioner’s challenge here. Even short of disavowal, we are also not persuaded that POSA would interpret glucose as excluded from possibly satisfying the “agent” limitation when other parts of the intrinsic evidence unambiguously list it as a membrane stabilizer and lysis inhibitor.²⁴

2. Has Petitioner established that the dextrose component of ACD impedes cell lysis in Landes’s method?

Landes describes processing maternal blood samples in Becton Dickinson’s ACDA collection tubes as part of a plasma separation protocol. *See, e.g.*, Ex. 1003, 10:26–32. It is, however, undisputed on this record that there is no express disclosure in Landes about the effects of using those ACD tubes on cell lysis specifically—much less what effect, if any, the dextrose component in those tubes has on cell lysis in Landes’s method. Petitioner argues that “the method of *Landes* anticipates even though it does not discuss the effect of glucose in the blood sample.” Pet. 26. According to Petitioner, the ’720 patent itself “states that glucose was a known ‘agent that stabilizes cell membranes’ and ‘reduce[s] maternal cell lysis.’” *Id.* at

²⁴ In the remarks where applicant unambiguously disclaimed EDTA, explaining that it was merely a “well-known chelator,” applicant contrasted EDTA with examples of inhibitors of cell lysis expressly listed in the patent, including glucose. Ex. 2064, 1196 (arguing that “EDTA is clearly referred to as a chelator in Applicant’s specification, . . . whereas multiple examples of agents that inhibit cell lysis are provided separately (see, e.g., paragraphs [0166] to [0167])”); *id.* at 1381 (paragraph 167, listing “glucose”).

27. Petitioner asserts that Landes is anticipatory regardless of whether Landes intended or recognized that glucose was a lysis-inhibiting agent. *Id.* at 26–28 (citing precedents on inherency); *see, e.g., Schering Corp.*, 339 F.3d at 1377 (holding that a “prior art reference may anticipate without disclosing a feature of the claimed invention if that missing characteristic is necessarily present, or inherent, in the single anticipating reference”); Pet. Reply 5 (arguing recognition of an inherent feature is not needed).

We agree with Patent Owner that a key issue here is whether “dextrose, when present in ACD under the conditions set forth in Landes, necessarily is the claimed agent.” PO Resp. 30. To be sure, the ’720 patent states that glucose is a membrane stabilizer that can be added to reduce cell lysis. And that statement is entitled to weight. But, as pointed out by Patent Owner, the patent “never states that ACD (or its components used in combination) disclosed in Landes is a membrane stabilizer that impedes cell lysis.” *Id.* at 32–33 (arguing that “[t]he *mere possibility* of ACD or its dextrose component impeding cell lysis is not sufficient to support inherency”); PO Sur-reply 6–7 (“Just because the ’720 patent identifies glucose as a membrane stabilizer does not mean that glucose under any and all conditions—or specifically in Landes’s ACD samples—necessarily impedes cell lysis.”).

In route to deciding this issue, we reject the contention that a mere presence of glucose in Landes’s sample is all that must be shown for Petitioner to prevail. As confirmed at oral argument, Petitioner urges an alleged “plain and ordinary meaning” of the claims. Tr. 19:16–20:3. But, in Petitioner’s view, it is simply a “capability” of glucose to inhibit lysis of

cells under some set of conditions that is needed—no specific level of inhibition, or even *any* inhibition of cell lysis, under Landes’s conditions is required. *Id.* at 18:7–12. By the plain language, however, claim 1 requires the sample include an “agent that impedes cell lysis, *if cells are present*,” and cells are doubtless present in the maternal blood samples taken in Landes’s methods. Petitioner cannot carry its burden simply by showing that glucose is introduced to Landes’s blood samples via ACD. Rather, Petitioner must show that glucose in Landes’s samples actually impedes at least some cell lysis, although we agree with Petitioner Landes itself need not have recognized that effect. We also agree with Patent Owner that the claims do not require “just any ‘agent’ . . . ; the claims require an agent ***that impedes/inhibits cell lysis***, albeit without requiring that agent to impose a specific amount of inhibition.” PO Sur-reply 8.

Petitioner contends we should accept as sufficient its mere presence-of-glucose position because Patent Owner allegedly charged others with infringement based solely on the addition of an “agent” (formaldehyde or formaldehyde-releasing compounds). Pet. Reply 6 (citing certain testimony of Drs. Chalmers and Van Ness). We decline to do so on this record. Whether some of Patent Owner’s witnesses may be applying the claims too broadly here or in other proceedings relative to infringement or objective indicia matters is not decisive on this issue because the intrinsic evidence and plain meaning of the claims controls.²⁵

²⁵ Similarly, the cited 2008 paper by Zhang (Ex. 2009, 1), which published many years after the application that issued as the ’720 patent was filed,

If some inhibition of cell lysis attributable to glucose in Landes's samples is required, Petitioner argues it has, nevertheless, made this showing. Tr. 22:10–22; Pet. Reply 8 (“Ravgen is incorrect that the glucose in the samples of *Landes* is not an agent that impedes cell lysis according to the patent claims.”). In support, Petitioner highlights two asserted properties of glucose that are alleged to prevent cell rupture: *isotonicity*, and providing a *nutrient source* to the cells. Pet. Reply 8. (citing, e.g., Ex. 1067, 59:5–20, 60:3–13, 75:24–76:8; Ex. 1043 ¶ 25; Ex. 2294, 213:2–214:17, 216:18–217:24; Ex. 1068 ¶¶ 2–7); Tr. 14:10–19 (Petitioner's counsel confirming reliance on glucose's two alleged functions: “isotonicity” and “nutrient source”).

Patent Owner contends these arguments by Petitioner about glucose's properties are untimely. PO Sur-reply 10–11. On this record, we do not agree. Isotonicity is discussed in Patent Owner's relied-upon evidence. *See e.g.*, Ex. 2074. We also find that the asserted properties of isotonicity and providing a potential nutrient source are fairly responsive to Patent Owner's argument that cells in the sample could not metabolize glucose under Landes's method, and that Petitioner has otherwise failed to show that glucose inherently inhibits lysis of cells as claimed. It was, thus, appropriate for Petitioner to address these matters further in its Reply. We address the alleged isotonicity and nutrient source properties in turn below.

Starting with isotonicity, Petitioner contends “that glucose ‘maintains

reporting “no effect” on cffDNA percentages for samples processed at six hours after collection in the presence of formaldehyde cannot change the meaning of the claims. Pet. Reply 7 (emphasis omitted).

the isotonicity’ of the sample in the presence of the citrate” in ACD. Pet. Reply 8 (quoting Ex. 1067, 60:3–13 (deposition testimony of Dr. Van Ness)); Ex. 1068 ¶ 3 (Dr. Levy noting glucose’s isotonicity property). According to Petitioner, “Dr. Van Ness confirmed that, ‘[i]n the absence of an isotonic solution cells have the potential to rupture.’” Pet. Reply 8 (quoting Ex. 1067, 59:5–20). Petitioner also cites to papers by Loutit (Ex. 1060) and Rapoport (Ex. 1062) as “compar[ing] citrate solution with acid citrate-glucose solution (ACD) and conclud[ing] that the addition of glucose improved cell stabilization.” Pet. Reply 8 (citing Ex. 1068 ¶¶ 4–6); Ex. 1068 ¶ 6 (Dr. Levy opining that “Rapoport explained that citrate creates a hypertonic solution, but with the addition of glucose, the solution ‘is actually almost isotonic with blood in terms of effective tonicity, owing to the permeability of the cellular membrane to glucose’”) (quoting Ex. 1062, 601–602)).

We agree that the cited evidence on isotonicity supports that glucose in ACD has the ability to inhibit cell rupture by helping to maintain fluid balance within the cells. Rapoport, for example, found that erythrocytes (RBCs) in an acid-citrate, glucose solution underwent osmotic hemolysis more slowly than RBCs in a simple citrate solution. Ex. 1062, 613. Patent Owner cites, however, rebuttal evidence, which we credit, that failure to maintain isotonicity does not *necessarily* lead to lysis. PO Sur-reply 11–12.

The failure to maintain tonicity causes cells to shrink or swell depending on the solution in which those cells are placed, but does not necessarily result in cell lysis. This is admitted by Dr. Levy on cross examination: “When a cell swells, it doesn’t always lead to lysis.” Ex. 2299,

31:22–32:12 (explaining it only increases the “potential” for lysis and “it’s only when the pressure on the membrane becomes I believe too great that you would get that lysis”); *id.* at 27:4–9 (discussing tonicity); Ex. 1068 ¶ 6 (opining that “without the glucose of ACD, blood cells in a sample would have a greater ***potential*** for cell lysis”) (emphasis added).

Dr. Levy also admits other variables can impact whether cells may lyse beyond tonicity. Ex. 2299, 33:19–25. Yet Dr. Levy did not consider such variables or conditions to show persuasively how the glucose of ACD under Landes’s conditions, would necessarily inhibit cell lysis in Landes’s method where samples are promptly processed within minutes and frozen. *Id.* at 37:19–22 (did not consider pH), 40:4–9 and 41:6–42:10 (did not consider temperatures); PO Resp. 34–36 (noting differences in, *inter alia*, processing times and temperatures, and glucose concentration compared to Landes); PO Sur-reply 9–12²⁶ (arguing the conditions of Petitioner’s references do not resemble Landes’s conditions); Ex. 1060, 3 (“Blood was stored . . . for 4 weeks” in Loutit); Ex. 1062, 594 (storage of whole blood

²⁶ Patent Owner suggests Dr. Levy did not consider glucose concentrations (PO Sur-reply 12 (citing Ex. 2299, 39:11–21)), but that is not entirely accurate. Dr. Levy opines that the glucose concentration of ACD in Landes is “comparable” to the glucose concentrations in some cited references. Pet. Reply 8–9; Ex. 1068 ¶ 7 (comparing 24.5 g/L of the Becton Dickinson ACDA tubes to the concentrations of 27.3 g/L and 30 g/L of Loutit and Rapoport (Ex. 1060, 186 and Ex. 1062, 593, respectively); *but see* Ex. 2299, 69:14–25 (Dr. Levy admitting, when asked whether he compared the concentrations of components in Rapoport’s solution to concentrations of ACDA (as in Landes) that he “didn’t look to draw an opinion on that”).

samples in a refrigerator at 4° C, followed by transport to a laboratory for testing over the course of days and weeks in Rapoport); Ex. 1003, 10:28–11:2, 16:20–24, 19:19–23 (Landes’s conditions where plasma is separated from the cellular fraction within about 20 minutes and promptly frozen at - 80 °C).

Turning next to glucose as a “nutrient source,” this is not taught in Landes and Petitioner has not shown that glucose necessarily provided a lysis-inhibiting nutrient source in Landes’s method. Petitioner’s evidence, including Mollison,²⁷ provides some support that glucose can be a nutrient source leading to increased RBC survival, and decreased cell breakdown and lysis, but we agree with Patent Owner and Dr. Van Ness that Mollison is discussing those effects upon longer-term storage of blood samples that are not applicable to Landes.²⁸ Ex. 1011, 16 (disclosing that glucose can also

²⁷ Mollison, P.L., *Historical Review: The Introduction of Citrate as an Anticoagulant for Transfusion and of Glucose as a Red Cell Preservative*, British J. of Hematology, 108:13–19 (2000) (Ex. 1011 or “Mollison”). Mollison provides a historical review of the development of citrate and citrate glucose compositions arising in connection with blood transfusions and experiences during World War I and afterward. *See generally* Ex. 1011 (discussing citrate as an anticoagulant and glucose as a preservative: “[T]he use of citrate made it possible to separate donor and recipient in space; the use of glucose made it possible to separate donation and transfusion by a substantial period of time.”).

²⁸ Petitioner cites several references as purporting to show glucose’s “known” properties. Pet. 16 (citing, e.g., Ex. 1011 (Mollison), Ex. 1019 (Lee), Ex. 1022 (Carter)). But Petitioner’s citation is conclusory, providing no substantive, supporting explanation pertinent to the present challenge. Dr. Levy’s corresponding testimony on this point cited by Petitioner is equally vague and unhelpful. Ex. 1056 ¶ 27.

“support red cell metabolism during storage”); Ex. 1067, 75:24–76:5 (Van Ness testimony that “[g]lucose can be a nutrient for cells”); PO Resp. 35 (citing Mollison’s teaching that samples were stored for weeks); Ex. 2239 ¶ 131 (Van Ness’s testimony that “Mollison’s disclosure is not applicable to . . . Landes, where cells are not stored . . . over a four week span”).

Relevant to Petitioner’s “nutrient source” theory, another condition, besides time, is temperature. On cross-examination, Dr. Levy testified that “it’s possible” cells “would be able to metabolize glucose [even] when they were stored at 3 to 7 degrees C.” Ex. 2299, 62:15–25. Dr. Levy further testified that “the colder you get, the slower the biochemical reactions occur” (*id.*) but ultimately conceded that “I didn’t look at these things or was asked to consider these things for . . . refuting or discussing the issues that I disagreed with Dr. Van Ness” (*id.* at 63:4–7); *see also id.* at 64:2–16 (admitting “I wasn’t asked to consider temperatures” on the question of whether cells are nonfunctional at low storage temperatures). Petitioner provides no persuasive evidence to indicate that any cells in Landes’s samples were impeded from undergoing lysis by virtue of cells metabolizing glucose under Landes’s conditions. Quite the opposite, Dr. Levy admitted he could not say if any cells in Landes’s method used glucose as a nutrient and avoided lysis as a result. *Id.* at 59:25–60:7 (“Q. Do you know if the cells would have used that glucose and metabolized it as a nutrient source in the diluents 1 to 8 [of Landes]? A. That wasn’t the focus of what I looked into, and that wasn’t directly measured in this article.”).

Patent Owner, on the other hand, cites evidence that cells are incapable of using glucose as a nutrient source under certain conditions—

especially at low temperatures. PO Resp. 29–30. Noting McMillan (Ex. 2235, 662–665) and its disclosure that “[r]apid separation of the sample or cooling will also prevent glycolysis [(i.e., cellular metabolism of glucose)],” Dr. Van Ness opines that the POSA would understand that “cells collected by Landes did not—and could not—metabolize or use the dextrose component of ACD as a membrane stabilizer or cell lysis inhibitor.”

Ex. 2239 ¶¶ 126–127. That is so, Dr. Van Ness explains, because Landes’s method rapidly processes the blood to separate plasma after which the plasma is transferred to a fresh tube and promptly frozen at -80°C. *Id.* ¶¶ 124–125. And Dr. Levy agrees that, from Landes’s disclosure, this process takes only “twenty plus a couple of minutes.” *Id.* (citing Ex. 2294, 151:19–152:7). Petitioner dismisses McMillan because it analyzed the cells’ inability to metabolize glucose already present in the blood, not glucose separately added as part of ACD. Pet. Reply 9. But if the cells are rendered nonfunctional and unable to consume endogenous glucose as a nutrient source at low temperatures, as suggested by McMillan and Dr. Van Ness, Petitioner fails to provide evidence or persuasive technical reasoning to suggest the cells’ behavior would change if more glucose were added. Ex. 2239 ¶¶ 126–127; Ex. 1068 ¶ 8.

Altogether, considering the argument and evidence of record, Petitioner has shown, at best, that it is *possible* for glucose to help impede cell lysis by reason of isotonicity and providing a nutrient source. But such possibilities are not enough for Petitioner to prevail on its challenge to the claims as anticipated by Landes. *In re Oelrich*, 666 F.2d 578, 581 (CCPA 1981) (“Inherency, however, may not be established by probabilities or

possibilities. The mere fact that a certain thing *may* result from a given set of circumstances is not sufficient.”) (internal quotation and citation omitted); *Guangdong Alison Hi-Tech*, 936 F.3d at 1364 (same). Petitioner has not, on this record, met its burden to show that the glucose component of ACD in Landes’s specific method is an “agent that impedes cell lysis” in the manner claimed.

Lastly, we address the argument about the comparative effects of EDTA and ACD. In its Response, Patent Owner cites a handful of references reporting no significant differences between use of EDTA and ACD as anticoagulants to prepare plasma samples as support for the proposition that the dextrose of ACD has not been shown to inhibit lysis of cells. PO Resp. 28–29 (citing, e.g., Ex. 1019, 5 (finding “no significant difference between EDTA and ACD plasma”); Ex. 2237, 2 (reporting “no statistically significant difference between log₁₀ copy numbers in samples collected in ACD tubes and those samples collected in EDTA tubes”)). Petitioner responds that this is an improper attempt by Patent Owner to read a “significant” lysis-inhibiting effect into the claims. Pet. Reply 5–6 (asserting such reading is “indefinite”). We disagree that was Patent Owner’s purpose. Instead, we understand that Patent Owner sought to show that EDTA (without dextrose) and ACD (with a dextrose element) were materially indistinguishable in terms of the reported results as anticlotting agents. PO Resp. 29.

This argument and evidence about comparing results with EDTA and ACD is ultimately neutral to our analysis. We have not been directed to evidence that calls out any specific effect of dextrose in these references. As

discussed above, EDTA is a chelator that prevents coagulation of the blood. ACD is also an anticoagulant, but includes a different chelator compound (sodium citrate/citrate ion). The argument and comparisons here between EDTA and ACD, which provide substantially the same anticoagulant effects using different chelators, shed little light on whether and to what extent any lysis inhibition is attributable to glucose in ACD, much less whether glucose had such an effect in Landes's method. Put differently, this argument and evidence fail to show specific effects of dextrose in Landes's ACD samples separate and apart from ACD's anticoagulation function generally.

3. Conclusion on Ground 1

For the reasons discussed above, we find Petitioner has not established by a preponderance of the evidence that Landes discloses "an agent that impedes cell lysis, if cells are present" as recited in independent claim 1, and the claims depending therefrom. Accordingly, Petitioner has not established that claims 1, 3–8, 21, and 24 are anticipated by Landes.

F. Ground 2: Obviousness over Landes and Marx

Petitioner asserts that claim 22 would have been obvious over Landes and Marx. Pet. 33–36. The art is summarized above. Claim 22 ultimately depends from claim 1. Petitioner relies on the dextrose in Landes's ACD for satisfying the "agent" limitation incorporated in claim 22 by virtue of its dependence on claim 1. Pet. 33. For the reasons explained above under Ground 1, Petitioner has not established by a preponderance of the evidence that Landes discloses the "agent" limitation of claim 1. As Petitioner's

Ground 2 relies on Petitioner prevailing on Ground 1 and the threshold showing for claim 1, Ground 2 also fails here.²⁹

G. Ground 3: Anticipation by Yen-Lieberman

Petitioner asserts that claims 1, 3–6, and 24 are unpatentable as anticipated by Yen-Lieberman. Pet. 36–41. Petitioner contends Yen-Lieberman is prior art under § 102(b). *Id.* at 20.³⁰ With respect to the “agent that impedes cell lysis, if cells are present” limitation, Petitioner points to Yen-Lieberman’s disclosure that “the AMPLICOR CMV detects DNA from plasma prepared from blood collected in an ACD tube.” Pet. 38. According to Petitioner, “[t]his is a reference to Becton Dickinson’s ACD blood collection tubes, which contain dextrose (*i.e.*, glucose).” *Id.* Thus, Petitioner concludes, “[f]or the same reasons explained . . . above with respect to *Landes*, *Yen-Lieberman* discusses a method using glucose, which the ’720 patent provides is ‘an agent that impedes cell lysis.’” *Id.*

²⁹ We need not reach Patent Owner’s argument on objective indicia of nonobviousness. *See Hamilton Beach Brands, Inc. v. f’real Foods, LLC*, 908 F.3d 1328, 1343 (Fed. Cir. 2018) (holding “objective indicia of nonobviousness” “need not [be] address[ed]” because the court “affirm[ed] the Board’s findings regarding the failure of the prior art to teach or suggest all [claim] limitations”).

³⁰ Before institution, Patent Owner contested the sufficiency of Petitioner’s showing that Yen-Lieberman was publicly accessible as of the date on the face of the document. We disagreed that Petitioner’s showing was insufficient. Inst. Dec. 53–55. Patent Owner did not raise any issue during trial about Yen-Lieberman’s prior art status and, thus, has forfeited any argument that Yen-Lieberman is not prior art. On this record, we determine that Yen-Lieberman is § 102(b) prior art based on the reasoning in the Institution Decision, which we adopt here. Inst. Dec. 54–55.

Patent Owner argues that Yen-Lieberman “includes only one brief mention of ACD” and in that brief mention, “discloses only that ACD is an anticoagulant.” PO Resp. 40. According to Patent Owner, “Yen-Lieberman provides no disclosure that ACD (or its dextrose component) is a membrane stabilizer, cell lysis inhibitor, or crosslinker that impedes cell lysis.” *Id.* Nor, according to Patent Owner does the Petition provide evidence “to support an inherency theory that the dextrose component of ACD as used in Yen-Lieberman is necessarily the claimed agent.” *Id.* at 42. Patent Owner thus asserts:

Petitioner argues that the dextrose component of ACD in Yen-Lieberman must necessarily impede cell lysis, but Petitioner and its expert provide no analysis of how or why this must necessarily occur, and Yen-Lieberman does not provide any detail (e.g., the concentration of dextrose in ACD tubes, the volume of blood collected, the temperature at which samples were processed, and buffer, all of which could potentially change how ACD affects a blood sample) to support Petitioner’s conclusory allegations and assumptions.

Id. at 45–46. Moreover, Patent Owner contends that the “blood collection and plasma separation in Yen-Lieberman would have occurred promptly and the plasma frozen until further processing as was standard for anticoagulant-based plasma preparation at the time.” *Id.* at 46 (citing Ex. 1003, 10:28–11:2; Ex. 1033, 7:6–19; Ex. 2286; Ex. 2239 ¶ 160; Ex. 2235, 1). Patent Owner concedes that “the ’720 patent states that glucose is a membrane stabilizer,” but argues that it “never states that ACD (or its components used in combination) is a membrane stabilizer that impedes cell lysis under the conditions disclosed in Yen-Lieberman.” *Id.* at 47.

As discussed *supra* § II.E.2, claim 1 requires the sample include an “agent that impedes cell lysis, if cells are present,” and Petitioner cannot carry its burden simply by showing that glucose is introduced to a blood sample. Rather, Petitioner must show that the glucose actually impedes at least some unspecified amount of cell lysis. In the context of Yen-Lieberman, this means that Petitioner must show that the glucose present in ACD impedes some degree of cell lysis under the conditions disclosed in Yen-Lieberman.

Yen-Lieberman mentions ACD in only a single sentence, which reads as follows (with the preceding and following sentences included for context):

The AMPLICOR CMB Test (Roche Diagnostics Corporation, Indianapolis, IN) is commercially available for research use only. This test uses PCR nucleic acid amplification for qualitative detection of HCMV DNA in plasma separated from anticoagulated whole blood (EDTA or ACD anticoagulants). This test permits the simultaneous amplification of HCMV target DNA and HCMV internal control DNA.

Ex. 1033, 107. We agree with Patent Owner that this passage does not expressly disclose that “ACD (or its dextrose component) is a membrane stabilizer, cell lysis inhibitor, or crosslinker that impedes cell lysis.” PO Resp. 40. Petitioner does not direct us to, and we do not find, any other passage in Yen-Lieberman that expressly supports that ACD impedes cell lysis.

Having determined that Yen-Lieberman does not expressly disclose that ACD impedes cell lysis, we next consider whether the glucose component of Yen-Lieberman’s ACD would inherently have that effect. As

Dr. Van Ness explains, Yen-Lieberman provides no information on the conditions under which ACD is used:

Yen-Lieberman provides no details that would inform a POSA of the use and function of the dextrose component of ACD or how the purported claimed agent impedes cell lysis in the sample. Yen-Lieberman does not convey the type of ACD tubes, the concentration of dextrose in the ACD tubes, the volume of the blood collected, the temperature at which the sample was processed, any buffer added to the sample, or information on any other factors that would impact the effects of dextrose in the sample.

Ex. 2239 ¶ 157. We agree with Dr. Van Ness.

As discussed *supra* § II.E.2, the conditions under which glucose is present with cells can affect whether the glucose acts to impede cell lysis. Absent such information, we lack an evidentiary basis to conclude, for example, that sample processing took place at a temperature and over a time period sufficient for the nutrient provision and tonicity maintaining properties of glucose to have a lysis-impeding effect. *See supra* § II.E.2. In short, there is insufficient evidence to conclude that glucose necessarily impedes cell lysis under the conditions present in Yen-Lieberman.

Petitioner does not address the conditions under which ACD is used in Yen-Lieberman. *See generally*, Pet. 37–38 (addressing the “agent” limitation without addressing conditions in Yen-Lieberman); Pet. Reply 11 (same). Instead, Petitioner argues that Yen-Lieberman’s ACD meets the claimed “agent” limitation “[f]or the same reasons explained in Section X.A.1 above with respect to *Landes*.” Pet. 38; *see also*, Pet. Reply 11 (“For the same reasons explained in Section II.A above with respect to *Landes*, *Yen-Lieberman* discusses a method using glucose, which is an agent that

impedes cell lysis according to the claims”). For the reasons discussed *supra* § II.E.2, these reasons are insufficient for Petitioner to carry its burden with respect to the “agent” limitation.

Accordingly, we find Petitioner has not established by a preponderance of the evidence that Yen-Lieberman discloses “an agent that impedes cell lysis, if cells are present” as recited in independent claim 1, and the claims depending therefrom. Thus, we find that Petitioner has not established that claims 1, 3–6, and 24 are unpatentable as anticipated by Yen-Lieberman.

H. Ground 4: Obviousness over Yen-Lieberman and Gocke

Petitioner asserts that claims 7–9, 21, and 25–30 would have been obvious over Yen-Lieberman and Gocke. Pet. 41–51. The art is summarized above. Claims 7–9, 21, and 25–30 ultimately depend from claim 1. For the reasons explained above with respect to Ground 3, Petitioner has not established by a preponderance of the evidence that Yen-Lieberman discloses the “agent” limitation of claim 1. As Petitioner’s Ground 4 relies on Petitioner prevailing on Ground 3 and the threshold showing for claim 1, Ground 4 also fails here.

I. Ground 5: Obviousness over Yen-Lieberman, Gocke, and Marx

Petitioner asserts that claim 22 would have been obvious over Yen-Lieberman, Gocke, and Marx. Pet. 51–53. The art is summarized above. Claim 22 ultimately depends from claim 1. Petitioner relies on the dextrose in Yen-Lieberman’s ACD for satisfying the “agent” limitation incorporated in claim 22 by virtue of its dependence on claim 1. *Id.* For the reasons

explained above with respect to Ground 3, Petitioner has not established by a preponderance of the evidence that Yen-Lieberman discloses the “agent” limitation of claim 1. As Petitioner’s Ground 5 relies on Petitioner prevailing on Ground 3 and the threshold showing for claim 1, Ground 5 also fails here.

III. MOTIONS FOR PROTECTIVE ORDER AND TO SEAL

Patent Owner moves for entry of a stipulated protective order and for an order sealing portions of the Patent Owner Response (Paper 19), all of Exhibits 2170–2173, and portions of Exhibit 2080. *See* Paper 21 (attaching stipulated protective order as Appendix A). That motion is unopposed.

A party may move to seal confidential information including, *inter alia*, sensitive commercial information. Consolidated Patent Office Trial Practice Guide, 19 (Nov. 2019); 37 C.F.R. § 42.54. It is the movant’s burden to show good cause for sealing such information, and we balance the party’s asserted need for confidentiality with the strong public interest in open proceedings. *Argentum Pharms. LLC v. Alcon Research, Ltd.*, IPR2017-01053, Paper 27 at 4 (PTAB Jan. 19, 2018) (informative).

Patent Owner provides a sufficient explanation for sealing the relevant portions of the Patent Owner Response and the identified exhibits. Exhibits 2170–2173 are license or settlement agreements between Patent Owner and third parties setting forth, for example, payment terms and sales data. Patent Owner states that it produced those agreements with the permission of third parties on the condition that the agreements remain sealed. Paper 21, 3–4. The subject portions of the Patent Owner Response and Exhibit 2080 include discussions about those agreements. Patent Owner

has also provided public redacted versions of Exhibit 2080 and the Patent Owner Response (Paper 20) so the record may remain clear and reasonably open.

Patent Owner has established good cause for sealing Exhibits 2170–2173, portions of Ex. 2080, and the Patent Owner Response. The stipulated Protective Order includes minor changes from our default language. That Protective Order, Appendix A to Paper 21, is entered.

Petitioner also filed a Motion to Seal (Paper 28). Petitioner contends that portions of its Reply (Paper 29) and portions of Exhibit 1066 (deposition transcript of Paul K. Meyer) should be sealed because those papers include information that Patent Owner considers contain confidential business information (e.g., licensing practices). Paper 28, 1. Petitioner has also filed public redacted versions of the Petitioner Reply (Paper 30) and Exhibit 1066. We find that, Petitioner has shown good cause for granting its motion. Petitioner’s Reply (Paper 29) and Exhibit 1066 are sealed.

IV. CONCLUSION

On this record, Petitioner has not shown by a preponderance of the evidence that any of the challenged claims are unpatentable based on grounds advanced in the Petition.

In Summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1, 3–8, 21, 24	102(a), (e)	Landes		1, 3–8, 21, 24
22	103(a)	Landes, Marx		22
1, 3–6,	102(b)	Yen-Lieberman		1, 3–6, 24

24				
7–9, 21, 25–30	103(a)	Yen-Lieberman, Gocke		7–9, 21, 25– 30
22	103(a)	Yen-Lieberman, Gocke, Marx		22
Overall Outcome				1, 3–9, 21, 22, 24–30

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that the Petitioner has not proved by a preponderance of the evidence that claims 1, 3–9, 21, 22, and 24–30 are unpatentable;

FURTHER ORDERED that Patent Owner’s Motion for Entry of Stipulated Protective Order (Appendix A to Paper 20) and to Seal is *granted*;

FURTHER ORDERED that Petitioner’s Motion to Seal (Paper 28) is *granted*; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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